

Bring a Barrier, Leave With a Build

Teachers Teach Teachers · 2 October 2026

Name

Subject · Grade

THE IDEA

RECURSION

The same rule, applied to its own output.

A coastline. A pair of lungs. A brain cell. **Now, AI.**

Going round again doesn't guarantee better. It makes things fit.

If the machine recurses on knowledge, the value isn't in knowing.



Massachusetts Institute of Technology

“not an optional exercise”

Sally Kornbluth, President, MIT · August 2026

46% use an LLM every day

90% worry they're becoming over-reliant

25% think MIT is preparing them to use AI

The Tech survey of MIT undergraduates, fall 2025

SO WE TEACH

 **Thinking routines.**

 **Creativity.**

This is why we start with reflection: to find the problems worth solving.

ALGORITHMIC

A known procedure. Follow the steps, get the answer. **AI has eaten this one.**

HEURISTIC

No procedure. Rules of thumb, judgment, trial. You know you've arrived when you look at it.

Every hard problem in your classroom is on the right.

What problems do you want to solve?



The loop
Barrier → Sketch → Prompt → Build → Test
We run it twice: Flint, then Gemini.

1 Warm-up

Round 1 · Write the most ridiculous word you can think of.

Round 2 · Now write your real classroom problem. One sentence.

2 Barrier, not deficit **BARRIER**

DEFICIT
They can't organise their writing.

BARRIER
The organising has to happen in their head at the same time as the writing.

Same student. Different problem. Only one of them is yours to fix.

YOUR TURN

_____ can't _____

The _____ has to happen at the same time as _____

3 Sketch it **SKETCH**

90 seconds. Boxes and arrows.



1 What does the student see first?

2 What can they do on it?

3 What happens when they tap?

4 The prompt spine **PROMPT**

Fill it in, then paste it into the tool.

Build a single-screen tool

for a Grade _____ student who _____.

Show _____ first,

then let them _____.

Keep the language at _____. No sign-in.

5 Build 01 **Flint** **7 minutes** **BUILD**

One drives. One talks. Your v1 will be wrong. That's the point.

What we built

6 Test 01 **3 minutes** **TEST**

Swap devices. They use it cold. You say nothing. Watch where they hesitate.

Where they hesitated

7 What it couldn't do

With your partner: what did you want it to do that it wouldn't?

8 Push it: SCAMPER

Substitute Combine Adapt Modify Put to another use Eliminate Reverse

Go to your corner. Change one thing. ☐ Substitute ☐ Combine ☐ Eliminate ☐ Mine works, push it

9 Build 02 **Gemini** **8 minutes** **BUILD**

Same barrier. More ambition. Nobody finishes this. You're finding out if it's buildable.

What Gemini can: **Custom logic. Multi-step. Behaviour you design.**

What problem do you want to solve next?



You just ran this twice. Run it again Monday.

This is just the beginning.

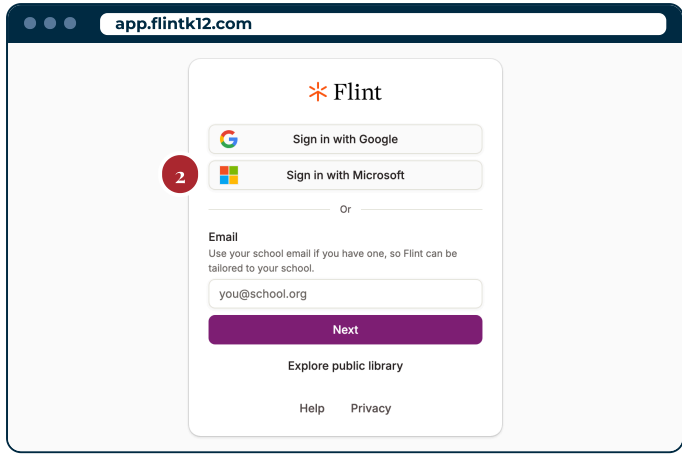
Keep working with your Technology Coach.

My coach _____

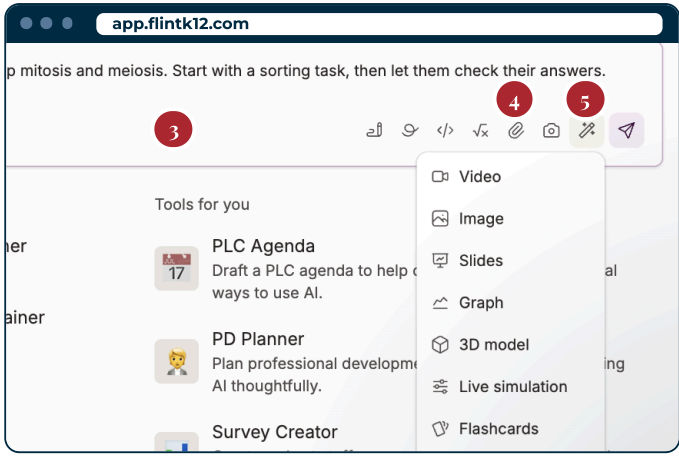


WORKSHOP PAGE

1 Log in and build



- 1 Go to **app.flintk12.com**
- 2 Sign in with **Microsoft**



- 3 Type your prompt in **How can I help you today?**
- 4 Add your materials with the **paperclip**
- 5 Press the **wand: Video or Live simulation**. Send.

2 Try a prompt

AN INTERACTIVE • LIVE SIMULATION

Create an interactive for Grade 9 students who mix up mitosis and meiosis. Start with a sorting task, then let them check their answers.

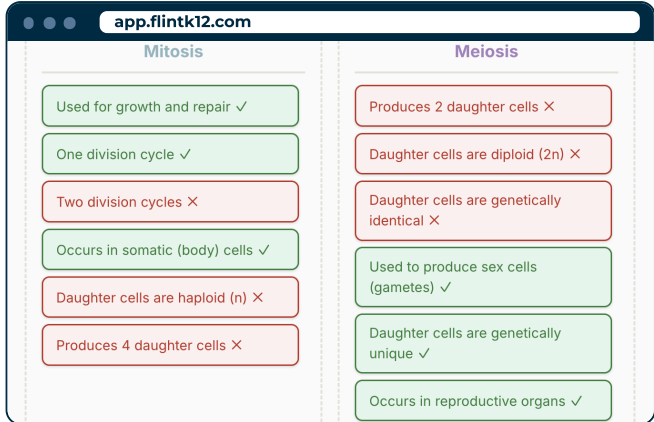
AN EXPLAINER • VIDEO

Make a two-minute explainer of how a bill becomes law, for Grade 8, using my attached slides. Pause after each step with one question.

3 What you can make

- Video
- Live simulation
- Slides
- Flashcards
- Practice problems
- 3D model
- Graph
- Image

Video makes an explainer. Live simulation makes an interactive.



Mitosis vs. Meiosis Sorting Challenge: made by Live simulation from the prompt on this page. Flint marks Live simulation *Experimental*.

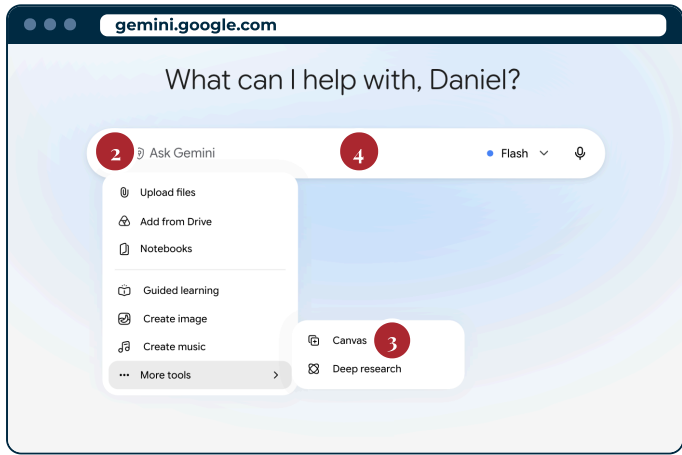
THEN PUSH IT, OR SHARE IT

- “Turn this into a Flint activity.” (for your class)
- “Add a teacher answer key.”
- “Make the challenge harder.”

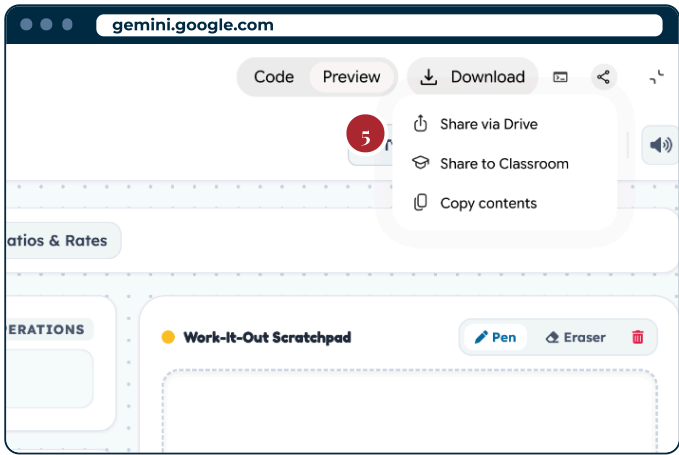
A GOOD PROMPT SAYS

who it's for · what they see first · what they can do · the language level · what to leave out. The prompt spine on the front does all five.

1 Log in and build



- 1 Go to **gemini.google.com**; sign in with **Google**
- 2 Press **+**
- 3 **More tools** → **Canvas**
- 4 Paste your prompt spine. Send.



- 5 Try it in **Preview**. Then **Share**: via Drive, or to Classroom
- TIP** Says it hit an error? Press **Try again**. Ours worked on the third try.

2 Try a prompt

THE PROMPT SPINE, FILLED IN

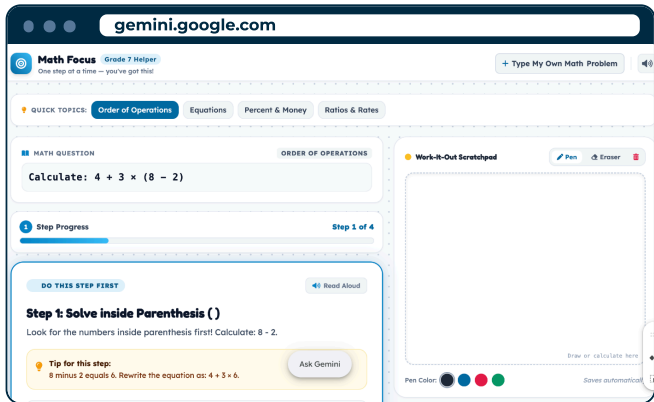
Build a single-screen tool for a Grade 7 student who loses their place in a multi-step maths problem. Show the current step first, then let them tick it off. Keep the language at Grade 5. No sign-in.

YOUR BARRIER

Paste your own prompt spine from the front of this sheet, then push it: one change at a time, like the SCAMPER corners.

3 What you can make

- A step-by-step scaffold for a task students lose track of
- A sorting or matching activity with instant feedback
- A self-check with hints instead of answers



Math Focus: made by Canvas from the prompt spine on this page

THEN PUSH IT

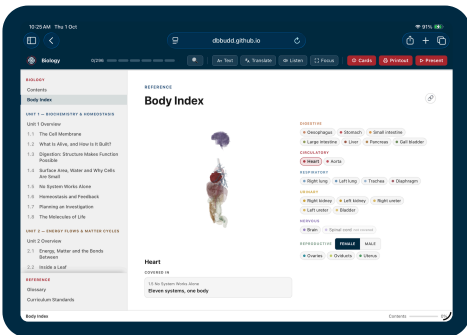
- “Make the buttons bigger for small hands.”
- “Add a reset button.”
- “Explain what each part does, in plain English.”

THREE WORDS

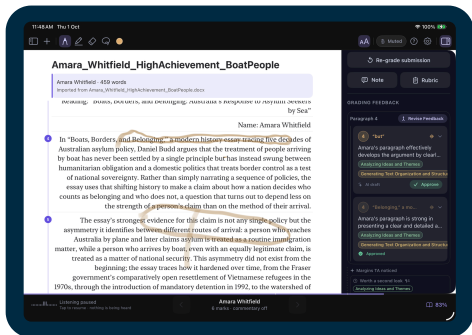
App Anything you open and use. It doesn't need to be in an App Store.
HTML The structure. Boxes, text, where things sit.
JavaScript The behaviour. What happens when you press the thing.

You write neither. You describe the behaviour.

The tool stops being the limit.
You become the limit.



Biology · dbbudd.github.io/biology



Margins · on-device AI feedback